



6-Month Project-Based Internship in Artificial Intelligence & Machine Learning

Program Objective

To build strong foundations and hands-on skills in AI and ML through real-world datasets, practical algorithms, deep learning, and end-to-end application development, enabling students to gain industry-ready expertise and a solid AI project portfolio.

Program Content

Pre-Phase (Bootcamp Weeks 0–1): Core Skills

Objective: Make sure everyone has the minimum foundation before starting ML.

Topics Covered:

1. Math Basics for ML

- Descriptive statistics
- Probability (basics, distributions)
- Linear algebra needed for ML (vectors, matrices, dot product)
- Calculus intuition (derivatives, gradients)

2. Python Fundamentals

- Variables, loops, functions
- Working with files
- Virtual environments, Jupyter, VS Code
- NumPy, Pandas basics

3. Data Analysis Foundations

- Understanding datasets
- Data types, missing values
- Basic visualizations
- Analysis patterns and insights

Mini Assignments:

- Quick stats exploration on a simple dataset
- 10 Python practice tasks
- A small Pandas-based analysis report

Phase 1: Foundations of AI & ML (Weeks 1–4)

Objective: Build essential skills required for AI/ML development.

Topics Covered:

- Introduction to AI, ML, and Deep Learning
- Python essentials for ML (NumPy, Pandas, Matplotlib)



- Data loading, cleaning, and preprocessing
- Exploratory Data Analysis (EDA)
- Feature engineering basics
- ML workflow & pipelines

Hands-on Projects:

- **Project 1:** Data Cleaning & EDA on Real-World Dataset
- **Project 2:** Basic ML Pipeline for a Prediction Task

Phase 2: Supervised & Unsupervised ML (Weeks 5–8)

Objective: Implement classical ML algorithms with real-world datasets.

Topics Covered:

- Regression: Linear/Ridge/Lasso
- Classification: Logistic Regression, Decision Trees, SVM, Random Forest
- Clustering: K-Means, Hierarchical
- Dimensionality reduction: PCA
- Hyperparameter tuning & evaluation metrics

Hands-on Projects:

- **Project 3:** House Price Prediction
- **Project 4:** Customer Segmentation
- **Project 5:** Classification Model with Feature Engineering

Phase 3: Deep Learning Foundations (Weeks 9–12)

Objective: Learn core deep learning concepts and build neural networks.

Topics Covered:

- Understanding Neural Networks
- Forward & backward propagation
- TensorFlow & PyTorch basics
- Building and training ANN/CNN models
- Regularization & optimization

Hands-on Projects:

- **Project 6:** Neural Network from Scratch (NumPy)
- **Project 7:** CNN-Based Image Classification (MNIST/CIFAR)

Phase 4: AI Specializations (Weeks 13–18)

Objective: Build domain-specific AI applications.

Track A: Natural Language Processing (NLP)

- Text preprocessing, embeddings
- RNNs, LSTMs, Transformers
- Sentiment analysis, text classification, summarization



Projects:

- NLP Project 1: Sentiment Analysis
- NLP Project 2: Text Summarization with Transformers

Track B: Computer Vision (CV)

- CNNs for visual tasks
- Data augmentation & regularization
- Transfer learning (ResNet, MobileNet, VGG)
- Object detection basics

Projects:

- CV Project 1: Image Classification with Transfer Learning
- CV Project 2: Mini Object Detection Project

Track C: Time Series & Forecasting

- Time series analysis techniques
- ARIMA/SARIMA/Prophet
- LSTMs for sequential data

Projects:

- Time Series Project 1: Sales/Demand Forecasting
- Time Series Project 2: LSTM-Based Prediction

Phase 5: AI System Development & Deployment (Weeks 19–22)

Objective: Learn to deploy models as usable applications.

Topics Covered:

- Model optimization
- MLOps fundamentals
- Model deployment using Flask/Streamlit
- Basics of cloud deployment (optional)

Hands-on Projects:

- **Project 8:** Deploy a ML Model as a Web App
- **Project 9:** Build an Interactive AI Application

Phase 6: Capstone Project (Weeks 23–24)

Objective: Complete an industry-style AI/ML project.

Key Tasks:

- Problem definition & data collection
- End-to-end ML pipeline implementation
- Model training, optimization & evaluation
- Deployment (optional)
- Documentation & final presentation



Program Outcomes

Upon completion, participants will:

- Build **8–12 hands-on AI/ML projects**
 - Understand **end-to-end ML workflows**
 - Get exposure to **NLP, CV, Deep Learning, and Deployment**
 - Develop an **industry-ready AI/ML portfolio**
 - Be prepared for roles like **AI Intern, ML Intern, Data Analyst, or Junior Data Scientist**
-